Step 2b - H4 - Microwave manufacturer design simulation

Tech baseline

Technologies used in the market

Full-wave 3D EM + thermal multiphysics suites (Ansys HFSS, CST Studio Suite, COMSOL RF+Heat Transfer, Altair FEKO)

How it works: FEM (HFSS, COMSOL) or FIT/FDTD (CST) solvers compute cavity fields; coupled thermal/CFD step converts dissipated power into temperature fields; engineers script parametric sweeps and optimization manually.

How well it performs: Experimentally validated against real applicators (e.g., 915 MHz single-mode applicator benchmark COMSOL vs CST); single runs minutes-hours; full design-space sweeps days-weeks on workstations

Limits: Built for antennas/PCB/radar, not heating: temperature-dependent dielectrics, moving loads and runaway need manual coupling; steep learning curve; ~$10-70k per seat; results highly sensitive to dielectric input data the OEM must measure itself

Microwave-heating-dedicated FDTD (QWED QuickWave + Basic Heating Module, QW-OptimiserPlus)

How it works: FDTD EM-thermal co-simulation with temperature-/enthalpy-dependent media, moving/rotating load support, source-frequency tuning and built-in optimizer — the closest existing product to the H4 idea.

How well it performs: Purpose-built features cut model-building effort vs generalists; validated in high-temperature processing studies (e.g., zirconia sintering, runaway prediction); niche vendor scale

Limits: Small company with limited marketing reach, ecosystem and multiphysics breadth; still requires expert users and measured dielectric data; little traction outside the specialist MW-heating community

Analytical/mode-matching and reduced-order cavity models + in-house codes

How it works: Modal analysis of canonical applicator geometries, equivalent-circuit load models, and OEM-specific empirical design rules accumulated over decades.

How well it performs: Seconds per evaluation; adequate for early sizing of single-mode applicators and waveguide feeds; embedded in OEM know-how

Limits: Fails for multimode cavities, irregular loads and coupled thermal behavior — exactly where expensive mistakes happen; knowledge is tribal, undocumented, and non-transferable

Build-and-test prototyping (calorimetry, thermography, product trials)

How it works: The incumbent workflow: fabricate applicator variants, measure field/temperature indirectly, iterate in the pilot plant.

How well it performs: Highest fidelity for final validation; typical industrial applicator prototypes cost order ~$50-500k and add weeks-months per iteration

Limits: Slow, expensive, and blind to internal EM fields (only surface temperature/leakage measurable); limits the design space an OEM can explore; scales poorly with custom-machine business

AI/surrogate-assisted EM optimization

How it works: Train surrogate models on solver results to explore large design spaces; emerging feature in vendor roadmaps (Ansys optiSLang-type, CST+SIMULIA AI).

How well it performs: Market analyses report AI surrogates cutting multi-day solver campaigns to hours; adopted in antenna/radar design first

Limits: Needs hundreds-thousands of training simulations and curated dielectric data — expensive to bootstrap; no microwave-heating-specific surrogate libraries exist today

Dielectric property measurement (cavity perturbation, coaxial probe, resonant methods)

How it works: Laboratory characterization of complex permittivity vs temperature/frequency/moisture — the mandatory input for every heating simulation.

How well it performs: Well-established methods; published comparisons show strong sensitivity of simulated temperature to dielectric accuracy; ProSeed could own a proprietary materials database

Limits: Data is scattered, mostly unpublished, and must often be re-measured per customer material — a structural bottleneck no solver vendor solves

Why our solution can be better

Generalist suites sell OEMs an expensive, expert-only toolkit and leave each manufacturer to re-implement microwave-heating physics — dielectric databases, moving-load handling, calorimetric power balance, runaway prediction — on top of $50k seats. A domain-focused design tool can win by shipping applicator/cavity templates, a proprietary temperature-dependent dielectric library, and built-in ISM/emission checks, collapsing model-building time from engineer-weeks to days and making optimization loops practical for mid-size OEMs that cannot justify HFSS specialists. QWED's QuickWave proves a niche product can exist; ProSeed's advantage over QWED is owning real equipment design + process data, and offering it as tool-plus-service where incumbents sell seats alone.

Key assumptions

• Only ~15-25 established OEMs plus ~100-300 smaller builders exist globally, so reachable customers number in the low hundreds — a niche that caps license revenue but suits focused B2B sales (assumed; consistent with market reports naming ~10-20 key players)

• Generalist EM seats cost ~$10k (COMSOL base+RF ~EUR 7.7k) to ~$50-70k (HFSS) before the expert labor to use them (measured, public price lists/forums)

• OEM design today still relies heavily on iterative prototyping at ~$50-500k per applicator iteration (assumed, industry-typical for custom RF hardware)

• Accurate results require measured temperature-dependent dielectric data per product — whoever owns that data owns simulation accuracy (measured, applicator validation literature)

• A specialist tool must be ~5-10x faster to a manufacturable design than the generalist workflow to overcome switching inertia (assumed)

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